

Quantizer-aware geometry, energy concentration and range-law behavior. (a) All 8,064 non-BOS layer-27 down-input token projections in the frozen (v1,v2) basis, standardized separately, with unit free-direction projections shown at the same multiplier of 1.0. Their in-plane lengths are 0.015 for Hadamard and 1.000 for PrismQuant; direction cosines are distinct from token standard deviations. (b) Seven depth-spanning layers: 1, 5, 9, 13, 18, 22 and 27. Solid curves preserve the original BOS-excluded rank-256 spectra at layers 1/13/27. Dashed curves use existing all-token rank-64 spectra at layers 5/9/18/22 and stop at the measured endpoint; they are not continuations of the solid-curve protocol. The common all-token protocol for all seven layers is shown in the appendix. (c) All 2,520 activation, 280 V-cache and 112 multi-slot observations, with the unchanged pooled reference $y=0.059802+0.866515x$ ($R^2=0.86$); $x=\sqrt{1-f}$ and y is range relative to paired Hadamard. The gray dashed line is identity. (d) All 5,342 available Qwen3-30B-A3B-Base expert observations, excluded from the reference fit. Marker area is fixed; teal denotes $<2,048$ routed calibration tokens and blue $\geq 2,048$. Brown points connect medians in ten equal-count f bins; vertical intervals show Q1–Q3, not confidence intervals. The blue line is the unchanged reference from c; no MoE line is fitted for display. Within-MoE OLS association R^2 is 0.61 for the 4,955 experts reaching the original 256-row cap and 0.43 for all experts; transferred-reference predictive R^2 is 0.61/0.43, respectively. The full-row median y/x is 0.95. Removing the 263 experts with $f>0.5$ reduces x variance by 71.38% and within-subset R^2 to 0.19, illustrating range sensitivity. Cold-expert reference residuals have $3.82\times$ the hot-expert standard deviation. Their covariance uses the original layer-pooled shrinkage prior; this association does not isolate its causal effect. Expert evaluation rows come from their calibration capture, so cross-model holdout applies to the dense reference fit, not to expert calibration.